Software Test Plan (STP) - Bank Management System

Project: Bank Management System
Version: 1.0
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Date: 16-09-2026
Status: Draft

1. Introduction

This document defines the Software Test Plan for the Bank Management System (BMS) v1.0. It describes the testing objectives, scope, strategy, test environment, schedule, roles and responsibilities, risks, traceability, and reporting activities.

Scope: Testing covers customer and administrator authentication, account creation and management, deposit, withdrawal, balance-related banking operations, fund transfer, transaction records, credential management, input validation, security, performance, reliability, usability, maintainability, and portability.

References: Bank Management System SRS v1.0, Sections 4 and 5; Requirements Traceability Matrix (RTM).

Definitions: BMS (Bank Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), PIN (Personal Identification Number), UAT (User Acceptance Testing).

2. Test Items

- Authentication and Login module
- Customer Account Creation and Management module
- Customer Credential Management module
- Deposit module
- Withdrawal module
- Funds Transfer module
- Transaction and Audit Logging module
- Administrator and Maintenance module
- Input Validation and Error Handling
- Local Data Storage and Recovery
- User Interface and Application Portability

3. Features to be Tested

Features mapped to the Bank Management System SRS requirement IDs:

- BMS-F-001: Validate customer authentication credentials
- BMS-F-002: Lock account after three consecutive failed authentication attempts
- BMS-F-003: Change login credentials after current-credential verification
- BMS-F-004: Terminate session after inactivity or explicit logout
- BMS-F-005: Create new customer accounts
- BMS-F-010: Verify sufficient balance and withdrawal limit
- BMS-F-011: Process withdrawal and update transaction-related data
- BMS-F-020: Accept deposit amount
- BMS-F-021: Accept supported deposit
- BMS-F-023: Update balance after successful deposit
- BMS-F-024: Generate deposit transaction record
- BMS-F-030: Transfer funds between eligible accounts

- BMS-F-031: Verify transfer funds and limits
- BMS-F-032: Update source and destination balances
- BMS-F-040: Authorized maintenance/diagnostic functions
- BMS-F-041: Record system errors and events
- BMS-F-042: Provide authorized system monitoring information
- BMS-F-043: Notify authorized personnel of critical failures
- BMS-NF-001 to BMS-NF-005: Performance, reliability, usability, maintainability and portability
- BMS-SR-001 to BMS-SR-005: Credential protection, masking, lockout, transaction validation and audit logging

4. Features Not to be Tested

- Actual banking-network connectivity or integration with real financial institutions
- UPI, credit/debit card networks, loans, and other services explicitly outside the academic project scope
- External banking hardware or infrastructure not implemented in the project
- Production deployment and real customer financial transactions

5. Test Approach / Strategy

Levels:

- Unit testing of individual modules such as authentication, account management and transactions
- Integration testing between application modules and local data storage
- System testing of complete banking workflows
- Acceptance testing (UAT) against the SRS acceptance criteria

Types:

- Functional testing
- Regression testing
- Performance testing
- Reliability and data-integrity testing
- Usability testing
- Portability testing
- Security testing
- Input validation and negative testing

Entry Criteria: Stable build delivered, test data prepared, test environment available, and required modules ready for testing.

Exit Criteria: All planned test cases executed, all high-priority requirements verified, no critical defects open, and applicable acceptance criteria satisfied.

5.1 Security Validation

- Verify customer password/PIN is stored in hashed form and not in plaintext.
- Verify password/PIN input is masked on screen.
- Verify account lockout after three consecutive failed login attempts.
- Verify negative, zero and non-numeric transaction amounts are rejected.
- Verify every login attempt and transaction is recorded in the local audit log with timestamp, account ID and outcome.

6. Test Environment

Hardware: Standard personal computer/laptop with at least a dual-core processor, 4 GB RAM, 500 MB free storage, keyboard, mouse and monitor.

Software: Windows or Linux operating system, C/C++ compiler supporting C++17, standard C++ library, and the Bank Management System application.

Test Data: Dummy customer accounts, administrator credentials, valid/invalid transaction amounts, eligible and ineligible destination accounts, and sample transaction records.

Storage: Local files used by the application for account and transaction data.

7. Test Schedule

- Test case design: 17-Sep-2026
- Environment setup: 18-Sep-2026
- Test execution start: 19-Sep-2026
- Test execution end: 24-Sep-2026
- User Acceptance Testing (UAT): 25-Sep-2026 to 26-Sep-2026

8. Test Deliverables

- Software Test Plan
- Functional and non-functional test cases
- Security test cases
- Test data
- Test execution records
- Defect reports
- Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Assigned Team Member	Responsibility
QA Lead	Ianna Elizabeth Reni	Prepare the test plan, coordinate testing activities, review test coverage, and track overall test progress.
Test Engineer	Pranav Reddy Melachervu	Design and execute functional, non-functional, and security test cases; record test results and defects.
Developer	Vikas N	Support implementation fixes, reproduce defects, perform defect triage, and assist with technical verification.
Product Owner	Pratheek P Shettigar	Review acceptance criteria, validate project scope, review test results, and provide final readiness/sign-off recommendation.

10. Risks and Mitigation

Risk	Mitigation
Unstable build or incomplete modules	Use early smoke builds and prioritize high-priority requirement testing.
Local data corruption during testing	Use backup test data and verify pre/post-transaction states.
Incorrect transaction validation	Perform positive and negative tests for deposit, withdrawal and transfer amounts.
Authentication/security defects	Execute dedicated security tests for credentials, masking, lockout and audit logging.
Platform differences between Windows and Linux	Build and run the same source code on both supported platforms.

11. Assumptions & Dependencies

- The application build will be available before test execution.
- Dummy customer and administrator test data will be available.
- The local file-storage mechanism will be accessible during testing.
- All requirements to be tested are documented in the Bank Management System SRS.
- The project remains within the academic scope defined in the SRS.

12. Suspension & Resumption Criteria

Suspend testing if:

- The application cannot start or the test environment is unavailable.
- A blocking defect prevents execution of more than 30% of planned test cases.
- Account or transaction data becomes corrupted and prevents reliable testing.

Resume testing if:

- Blocking defects are resolved and verified.
- The test environment is stabilized.
- Valid test data is restored or recreated.

13. Test Case Management & Traceability

The RTM ensures mapping of Bank Management System requirements to their corresponding test cases. Each requirement in Sections 4 and 5 shall be linked to one or more verification activities.

- BMS-F-001 (Authentication) → TC-Auth-01
- BMS-F-002 (Account lockout) → TC-Auth-03
- BMS-F-003 (Credential change) → TC-Auth-04
- BMS-F-004 (Logout/session timeout) → TC-Auth-05
- BMS-F-005 (Account creation) → TC-Login-01
- BMS-F-010/BMS-F-011 (Withdrawal) → TC-WD-01, TC-WD-02
- BMS-F-020/BMS-F-021/BMS-F-023/BMS-F-024 (Deposit) → TC-DEP-01, TC-DEP-02, TC-DEP-04, TC-DEP-05
- BMS-F-030/BMS-F-031/BMS-F-032 (Funds Transfer) → TC-TR-01, TC-TR-02, TC-TR-03
- BMS-NF-001 (Performance) → TC-Perf-01
- BMS-NF-002 (Reliability) → TC-Rel-01
- BMS-NF-003 (Usability) → TC-UX-01
- BMS-NF-004 (Maintainability) → TC-Maint-01

- BMS-NF-005 (Portability) → TC-Port-01
- BMS-SR-001 to BMS-SR-005 (Security) → TC-Sec-01 to TC-Sec-05

14. Test Metrics & Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage of test cases passed/failed
- Requirement coverage
- Number and severity of defects
- Defect aging
- Percentage of high-priority requirements verified

Reports:

- Test execution status
- Defect reports
- Requirements coverage report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Ianna Elizabeth Reni	
Test Engineer	Pranav Reddy Melachervu	
Developer	Vikas N	
Product Owner	Pratheek P Shettigar	